

## **Self-Assembled Antifreeze Peptides: A Biomimetic Route to Cryopreservation**

Peptide self-assembly offers a powerful strategy for biomaterials that mimic natural systems, including antifreeze proteins (AFPs), whose precisely arranged binding domains control ice crystal formation but whose extraction and instability limit practical use. We design short peptides that self-assemble into nanostructures mimicking AFP ice-binding mechanisms, where collective arrangement of ice-binding moieties amplifies interaction with ice and enhances recrystallization inhibition beyond individual molecules. Organic-inorganic nanohybrids further enable size- and shape-tunable control of ice-water interfaces, with optical properties allowing direct visualization of inhibition and mechanistic insight, yielding assemblies with strong cryoprotective efficacy and low cytotoxicity versus DMSO.

**Keywords:** peptide self-assembly, ice-binding proteins, cryopreservation, supramolecular nanomaterials, ice recrystallization inhibition, biomimetic materials